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Effects of Prescription Drug Insurance on Hospitalization and Mortality: Evidence From Medicare Part D

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Abstract

We used Medicare administrative data (2002–2009) and an instrumental variables design that exploits the natural experiment created by the implementation of Medicare Part D to estimate the effect of prescription drug coverage insurance on the use and costs of inpatient services. We find that gaining prescription drug insurance through Part D caused approximately a 4 percent decrease in hospital admission rate, a 2–5 percent decrease in Medicare inpatient payments per person, and a 10–15 percent decrease in inpatient charges. Among specific types of admissions, gaining insurance was associated with significant decreases in admissions for CHF and COPD.

Introduction

Congress created the Medicare Part D program to provide subsidized prescription drug insurance for Medicare recipients. The creation of Part D was motivated, in part, by the growing importance of prescription drugs in preventing and treating diseases, and the growing costs of buying prescription drugs.[1] As a result of Part D, the proportion of elderly with prescription drug insurance increased from approximately 66 to 90 percent (Kaiser Family Foundation, 2010; Levy and Weir, 2010; Engelhardt and Gruber, 2011; Kaestner and Khan, 2012; Zimmer, 2012).

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[1]For the 20 years prior to passage of Medicare Part D, prescription drug use was growing faster than the population and prescription drug spending was growing faster than overall spending on health care. See "Prescription Drug Trends," Kaiser Family Foundation (2010), http://www.kff.org/rxdrugs/upload/3057-08.pdf (accessed April 2, 2013).

The cost of Medicare Part D is substantial. Federal and State expenditures on Part D were $83.8 billion in 2014.[2] Moreover, the Congressional Budget Office (CBO) forecasted that expenditures on Part D will increase by $51 billion from 2013 to 2022 because of the expansion of the generosity of Medicare Part D contained in the Affordable Care Act that will gradually eliminate the gap in coverage popularly referred to as the doughnut hole (CBO, 2012). Notably, the CBO forecast was net of $35 billion in savings ("offsets") from Part D due to reductions in the use of other medical services because of greater use of prescription drugs. Without this offset, the forecasted increase in spending on Medicare Part D would have been $86 billion, or 68 percent greater. Indeed, during the debate over Part D, many legislators made statements reflecting the notion that obtaining prescription drug insurance will reduce other types of health care spending. Typical of such statements are the following taken from the Congressional Record June 17, 2003.[3]

> Now we find through research funded by Government, through research funded by the drug companies, and products that have emerged from that research, that many of the sicknesses you used to go to the hospital for and stayed for 3 or 4 days can be taken care of by taking a pill. Yet Medicare says if you go to the hospital and run up a bill of however many tens of thousands of dollars to stay that many days, we will pay for it. But if you take the pill that makes the hospital visit unnecessary, we will not. That clearly doesn't make sense. There is the need for the benefit of prescription drugs, and the Medicare system needs to catch up to circumstance. (Senator Robert Bennett, Republican, Utah)

> There is a dramatic change in the pattern and practice of medicine. Perhaps no better example is what happens with stomach illness. Twenty years ago, there was not much one could do for somebody who suffered from ulcers other than to have surgery. But now with prescription drugs that address the underlying causes, stomach surgery has been reduced by two-thirds. Yet, in Medicare there is no coverage for those prescription drugs. You can't have a modern Medicare without a prescription drug component. (Senator Kent Conrad, Democrat, North Dakota)

While Senators Bennett and Conrad, among others, believed that a prescription drug benefit would reduce spending on nonpharmacy services and improve health, the evidence at the time to support their statements was, at best, sparse. Moreover, there

[2]"2015 Annual Report of the Board of Trustees," Federal Hospital Insurance and Federal Supplementary Medical Insurance Trust Funds, https://www.cms.gov/Research-Statistics-Data-and-Systems/Statistics-Trends-and-Reports/ReportsTrustFunds/Downloads/TR2015.pdf (accessed February 10, 2016).

[3]See http://thomas.loc.gov/cgi-in/query/D?r108:12:./temp/~r108FHAhy2::.

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are plausible reasons to expect that prescription drug coverage would have little effect on nonpharmacy spending, for example, because of the possibility that much of the additional prescription drug use that comes with insurance may be marginal in terms of health benefits. The relative dearth of evidence on whether prescription drugs are substitutes for other medical spending persists today and is reflected in the recent CBO (2012) report, which relied on a limited number of studies to draw its conclusions about offsetting effects of Part D on spending on other medical services.[4]

In this article, we examine whether obtaining prescription drug coverage through the Medicare Part D program affected hospital admissions and inpatient expenditures associated with those admissions for a variety of illnesses such as congestive heart failure (CHF), stroke, dehydration, and chronic obstructive pulmonary disease (COPD). We also assess whether gaining Medicare Part D coverage affected all-cause mortality. Data for the analysis are drawn primarily from Medicare claims for inpatient services (Medicare Provider Analysis and Review, MEDPAR) from 2002 to 2009, thus spanning the implementation of Medicare Part D in 2006. We also use data from the Medicare Current Beneficiary Survey (MCBS) over the same period to obtain information about prescription drug insurance of Medicare beneficiaries because such information is not available in the data on inpatient claims. With these data, we compare hospital admissions, inpatient spending, and mortality before and after implementation of Medicare Part D for elderly who were more or less likely to gain prescription drug insurance through Medicare Part D.

We find that obtaining prescription drug insurance through Medicare Part D significantly reduced hospital admissions and Medicare expenditures for those admissions. Overall, gaining prescription drug insurance through Medicare Part D caused approximately a 4 percent decrease in the number of hospital admissions per 1,000 people, a 2 percent (full sample) to 5 percent (white-only sample) decrease in Medicare payments per person for hospital admissions, and a 10 percent (full sample) to 15 percent (white-only sample) decrease in inpatient charges. Among specific types of admissions, prescription drug insurance was associated with significant decreases in admissions for CHF, COPD, and diabetes. Associations between prescription drug insurance and resource use (i.e., hospital charges) were usually larger than associations between prescription drug coverage and the number of admissions, which implies that gaining prescription drug insurance affected resource-intensive admissions more than low-cost admissions. Gaining prescription drug coverage through Medicare Part D did not affect all-cause mortality. Given that approximately 11 million persons gained prescription drug insurance through Medicare Part D, mostly through stand-alone Part D plans, we estimate that reduced hospital expenditures associated with increased prescription drug coverage produced aggregate savings of approximately $500 million per year.

[4]The applicability of some of the studies reviewed in the CBO report to Medicare Part D is debatable. For example, Gaynor, Li, and Vogt (2007) examine nonelderly, and Stuart, Doshi, and Terza (2009) treat prescription drug coverage as exogenous. We review most of the other studies contained in CBO report below.

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RELEVANT LITERATURE

Prescription Drug Insurance and Hospitalization[5]

Relatively few studies have examined the association between prescription drug insurance and the use of nonpharmacy services, including hospitalization, among the elderly. The lack of research on these important questions stands in contrast to the large literature that has examined the effect of prescription drug insurance (or changes in cost sharing) on prescription drug use. In a systematic review of the literature, Goldman, Joyce, and Zheng (2007) conclude that there are a relatively small number of studies of the association between prescription drug insurance (or cost sharing) and health and use of other (e.g., inpatient) services. Moreover, these authors note that studies in this area usually focused on small samples of chronically ill patients. For the chronically ill, Goldman , Joyce, and Zheng (2007) report that most studies find consistent evidence that prescription drug insurance is associated with improved health. However, results from studies of broader populations are inconsistent. A similar conclusion is reached by Gibson, Ozminkowski, and Goetzel (2005) in their review.

There are several studies closely related to the current research, but instead of reviewing them individually, we summarize the gaps in knowledge that remain so as to highlight the contribution of our research. First, few studies use nationally representative samples of elderly persons. Soumerai et al. (1991), Johnson et al. (1997), Hsu et al. (2006), and Zhang et al. (2009) all use samples from one specific insurance plan to study how changes in prescription drug benefit design (Soumerai et al., 1991; Johnson et al., 1997; Hsu et al., 2006) or prescription drug insurance (Zhang et al., 2009) affect nonpharmacy medical spending. While these samples and settings provide natural experiments that are an advantage from a research design (i.e., internal validity) perspective, the narrowness of the samples and contexts limits the applicability (i.e., external validity) of the findings. For example, estimates of the effect of a change in prescription plan benefit generosity on hospitalizations of Medicare managed care enrollees in Northern California (Hsu et al., 2006) or managed care enrollees in Pennsylvania (Zhang et al., 2009) may not generalize to broader population and the effect of gaining prescription drug insurance through Medicare Part D.

Second, a considerable amount of previous research focuses on the effects of changes in benefit generosity rather than gaining prescription drug insurance (Soumerai et al., 1991; Johnson et al., 1997; Hsu et al., 2006; Chandra, Gruber, and McKnight, 2010; McWilliams, Zaslavsky, and Huskamp, 2011). It is unclear whether these studies can inform the debate over the possible offsetting effects of gaining prescription drug

[5]We limit the review to studies of elderly, so we exclude studies such as Gaynor , Li, and Vogt (2007), and we include studies that use methods that account in some way for the nonrandom choice of prescription drug insurance. We do not review studies of the effect of prescription drug insurance on elderly persons with specific illnesses (e.g., Soumerai et al., 1994; Pilote et al., 2002; Stuart et al., 2004; Hsu et al., 2006; Tjia and Briesacher, 2008). We also do not include studies of the effect of entering the coverage gap in Part D (see Stubbings and Lau, 2013, for a review).

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coverage through Medicare Part D because gaining prescription coverage represents a larger treatment, and extrapolating from these studies to infer the effect of Part D may be misleading.

Third, because of the non- and quasi-experimental nature of previous studies, there are standard concerns with respect to internal validity and additional research using different samples and methods is warranted.

Studies closest to ours are Afendulis et al. (2011), Liu et al. (2011), and Kaestner and Khan (2012). These studies examine the effect of gaining prescription drug insurance through Medicare Part D on the use of inpatient services for a representative sample of the Medicare population. They report mixed findings. Afendulis et al. (2011) and Liu et al. (2011) use a difference-in-differences approach with nonelderly as a comparison group for the elderly. Kaestner and Khan (2012) use an approach similar to the one we adopt. One concern with the difference-in-differences approach is whether nonelderly are a valid comparison for the elderly. As reported in Afendulis et al. (2011) and Dunn and Shapiro (2015), there were differences in rates of hospitalization in the preperiod between the elderly and nonelderly and differences in the trend in rates of hospitalization between the elderly and nonelderly. These differences suggest that the near-elderly may not be an appropriate comparison group for the elderly. A weakness of Kahn and Kaestner (2012) is statistical power because they use the MCBS survey, which has relatively small samples.

Prescription Drug Insurance and Mortality

We also assess how obtaining prescription drug insurance through Medicare Part D has affected all-cause mortality. Two studies examine similar questions. Huh and Reif (2017) study the effect of Medicare Part D on mortality for 66-year-olds using data from National Vital Statistics (2001–2008). They use an event study approach to detect any break in the trend of the difference in mortality between 66-year-olds and 64-year-olds around the implementation of Medicare Part D and a simple two-period difference-in-differences to obtain the estimate for the effect of Part D on mortality. They find a 2.2 percent reduction in annual mortality for 66-year-olds following Part D, and the reduction is primarily due to fewer cardiovascular deaths. Dunn and Shapiro (2015) examine how Part D affects cardiovascular-related and non-cardiovascular-related mortality for the elderly population (ages 65–84) using mortality data from the Centers for Disease Control and Prevention (2000–2010) and data on prescription insurance coverage from the Medicare Current Beneficiary Survey (years 2001–2009). They used an estimation strategy similar to the current approach (Kaestner and Khan 2012) and find that Medicare Part D decreases cardiovascular-related deaths, but the effect is small and not consistent across analyses. Notably, Dunn and Shapiro (2015) present evidence that Part D affects the nonelderly and that using the nonelderly as a comparison group, as in Huh and Reif (2017), may bias estimates.

Summary of Past Research

As our brief review notes, there is relatively little research on how prescription drug insurance affects use of other medical services and even less focused on Medicare Part

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D. Moreover, the literature has reported mixed findings. There are only two studies of the effect of Medicare Part D on mortality, and here too, there is inconsistent evidence. We add to this literature relatively small literature by studying how gaining prescription drug insurance through Medicare Part D affects several types of hospitalizations and mortality for a broad segment of the Medicare population.

DATA

The data we use are from three sources: the Medicare Provider Analysis and Review file (MEDPAR), the Medicare Master Beneficiary Summary File (MBSF or denominator file), and the Medicare Current Beneficiary Survey (MCBS). The MEDPAR files provide information on all hospital admissions for Medicare beneficiaries not enrolled in Medicare Advantage (i.e., capitated HMO). The Denominator file provides information on all Medicare beneficiaries and the reason for their eligibility, whether they are in Medicare Advantage plan, whether they have prescription drug insurance (2006 onward), whether they are disabled or have ESRD, and the year of death. The MCBS provides information on prescription drug insurance. The years covered by each of these data sources are 2002–2009.

We focus on persons 66–84 years old on December 31 of the year and enrolled in fee-for-service Medicare. We exclude people in Medicare Advantage because they do not have information in MEDPAR claims. We also exclude people covered by Medicaid and people with ESRD or Disability because they are quite sick relative to the large majority of Medicare enrollees, and their greater level of illness may cause them to have significantly different trends in hospitalization and mortality. The sample we use covers approximately 65 percent of the Medicare population. In some analyses, we restrict the sample to white persons. The sample (from MCBS) of nonwhites is too small to analyze separately. For the analysis of hospitalizations, we exclude people who died because they did not have full-year information, but we conduct an analysis on whether mortality is associated with prescription drug insurance. Finally, we drop the year 2006 because that was the year of implementation, although results are similar when 2006 was included.

The main limitation of the Medicare administrative data for our purposes is that they do not provide information on prescription drug insurance prior to 2006. To overcome this limitation, we used the MCBS, which is a nationally representative survey of Medicare beneficiaries that has prescription drug insurance information, to estimate prescription drug insurance prior to 2006 for Medicare beneficiaries in our sample. Individuals in the MCBS are drawn using stratified random sampling from an enrollment list of persons entitled to Medicare on January 1 of that year, and the survey is intended to be representative of all geographical areas and age groups. Each year, a supplemental sample is added to account for attrition so as to maintain an average sample size of approximately 12,500 individuals. Persons remain in the sample for 4 years. We use the Cost and Use file of the MCBS, which provides detailed information of the respondent's insurance status including prescription insurance and demographic and socioeconomic characteristics.

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In the MCBS Cost and Use file, respondents report up to five sources of non-Medicare insurance and prescription drug coverage. The respondents report the start and stop dates of the insurance and whether this source of insurance provides prescription drug coverage. Based on this information, we assign a person to the insured or uninsured category for each month in the survey. We define someone as uninsured if they did not have prescription drug insurance for the full year.[6] After constructing the measure of prescription drug insurance for each person, we use sample weights and aggregate the MCBS data into demographic cells based on two periods (2002–2005, 2007–2009), eight age groups (66–68, 69–70, 71–72, 73–74, 75–77, 78–79, 80–81, and 82–84), two gender groups, and nine Census divisions.

We aggregate the MCBS data across years (e.g., 2002–2005) because of relatively small sample sizes to reduce measurement error in estimates of prescription drug insurance, and we use these demographic characteristics because information about these characteristics are available in the Medicare administrative data, which allows us to merge the prescription drug insurance information from MCBS with the MEDPAR data.[7] Aggregation yields 230 cells: 115 age-gender-region cells for each of two periods. While there are 288 possible combinations of period (2), age (8), gender (2), and Census division (9), we use 230 because we select cells with 75 or more observations in the MCBS. We tried different thresholds, for example, 100, and results are similar. Note that we do not have different values of prescription drug insurance for each cell-year. So, when we merge the data to the MEDPAR data, we assign the same value of prescription drug insurance of a cell for each year in the two periods: 2002–2005 and 2007–2009.

We also aggregate the administrative data by the same 115 demographic cells defined by age-gender-Census division. The Denominator file provides the number of persons covered by Medicare in each demographic cell defined by year, age, gender, and Census division, which we use as the denominator in the calculation of cell hospitalization rates. The Denominator file also provides information on the year of death, which we use to construct rates of all-cause mortality (per 1,000 persons). We apply the same sample selection criteria to these data. There are a total of 805 cells reflecting the 115 demographic groups and 7 years of administrative data.

We also use the Denominator file to construct the proportion of people in each demographic cell enrolled in Medicare Advantage. Because we do not include persons enrolled in Medicare Advantage in the analysis and because the proportion of enrollees was increasing during our period of analysis, we construct the proportion of persons enrolled in Medicare Advantage to include in the regression models to account for any potential changes in sample composition that could be correlated

[6]Approximately 90 percent of persons who were uninsured at least 7 or more months during the year were uninsured for the entire year. Using a less stringent definition (e.g., 7 or more months uninsured) has little effect on estimates.

[7]Ideally, we would like to use more characteristics than these three, but as we describe below, the sample size of the MCBS limits the amount of characteristics we can use. For example, the MCBS is too small to use state instead of region, and too small to use race. However, the variables we use are effective, as we demonstrate below.

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with the implementation of Medicare Part D. Including or excluding this variable makes little difference to results reported below.

The MEDPAR file provides information on all hospitalizations for the sample of Medicare beneficiaries. We sum the number of hospitalizations in each of the 115 demographic cells in each year and this count becomes the numerator in the cell-specific hospitalization rates. Note that hospitalization rates are measured per 1,000 persons. We sum hospitalizations within a cell in two ways: first, by the number of people, so persons with more than one hospitalization are only counted once, and second, by the number of hospitalizations, which counts all hospitalizations. Results do not differ significantly for the two measures so we only report results using the latter measure.

We define several measures of hospitalization. We select 8 of the top 10 causes of admission among elderly based on AHRQ analysis of HCUP data (Russo and Elixhauser, 2006). These are: CHF, pneumonia, coronary atherosclerosis, cardiac dysrhythmias, AMI, osteoarthritis, stroke, and COPD. These 8 are taken from a list of 10 and account for approximately one-third of all admissions. We exclude rehabilitation care, fluid and electrolyte disorders, and hip fracture because these 3 are not in top 10 in every year. Also, these 3 categories are relatively small individually and combined account for less than 3 percent of admissions in a year. We augment this list of 8 conditions with 5 (Prevention Quality Indicator (PQI) admissions from AHRQ that are not already included among our 8 most common admission types: dehydration, UTI, diabetes (4 subcategories aggregated), angina, and hypertension.[8] PQI admissions are deemed amenable to primary care including prescription drug use. We also include admissions for trauma, which consist of a relatively large group of conditions (e.g., burns, fractures). Because it is less likely that prescription drug insurance would affect trauma admissions, trauma admissions are used as a falsification test. Definitions of trauma hospitalizations are based on information from the Agency for Healthcare Quality and Research.

In addition to the number of hospitalizations per 1,000 persons, we also construct measures of expenditures for each type of hospitalization. We measure expenditures as the total Medicare payments made to the hospital for each type of admission per person. To construct expenditures, we sum all payments for a specific type of hospitalization for all persons in a cell and then divide by the number of persons in that cell. Therefore, this is a measure of expenditures per person for each type of hospitalization. We construct a similar variable using total hospital "charges" for the admission, which we view as a measure of resource use. We use hospital charges because Medicare payments do not necessarily reflect true resource costs, which is important from an economic point of view. Charges include all resources used, but at

[8]The definitions of hospitalizations are taken from the following two sources: http://www.qualityindicators.ahrq.gov/Modules/PQI_TechSpec.aspx. and "Statistical Brief #14" by HealthCare Cost and Utilization Project, October 2006. The link for the codebook is "Supplementary Appendix SA—Clinical Classification Software-DIAGNOSES (January 1980 through September 2013)" http://www.hcup-us.ahrq.gov/toolssoftware/ccs/AppendixASingleDX.txt.

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well-known inflated "prices" that vary across hospitals. However, for the estimates of the effect of insurance on charges to be confounded by changing "prices," charges would have to change differentially over time by demographic group and be correlated with changes in prescription drug insurance. While we cannot rule this possibility out, there is no obvious reason for it to be so. Nevertheless, this is a potential limitation of using charges, but here we note that estimates are similar in magnitude whether we use Medicare payments or charges.

Table 1 reports the weighted, sample means, and standard deviations for the different types of hospitalizations by selected years. We show means for the full sample and for a sample restricted to whites. The 13 individual types of admissions account for approximately 38 percent of all hospital admissions, which is consistent with the selection of the most common admissions. Approximately 25 percent of the sample is admitted to the hospital each year. Total admissions remained relatively constant from 2002 to 2006, but declined by approximately 10 percent between 2006 and 2009. Many admissions, particularly for heart disease and respiratory illness, declined substantially over the period. However, admissions for osteoarthritic conditions increased by approximately 25 percent. Mortality rates were 32 per 1,000 in 2002 and declined steadily to 27 per 1,000 by 2009. These figures are consistent with other published estimates.[9]

EMPIRICAL APPROACH

Conceptual Model

The empirical analysis is motivated by basic, economic theory and substantial empirical evidence that obtaining insurance for prescription drugs, for example through Medicare Part D, increases the use of prescription drugs because insurance lowers the price of prescription drugs (moral hazard effect) and increases the ability to pay for prescription drugs (income effect).[10] Because virtually all elderly in the United States had medical insurance through Medicare prior to Part D, elderly without prescription drug insurance may have underutilized drug therapy vis-à-vis nondrug treatments because medical treatments were heavily subsidized. However, after gaining prescription coverage through Part D, which substantial evidence shows has increased prescription drug use, the use of nonpharmacy services such as inpatient care may have decreased because of substitution of now cheaper drug therapy for medical treatment.[11]

Whether substituting drug therapy for medical treatment will improve or worsen health depends on whether drug therapy is more or less efficacious than the medical treatments it replaced. It may be rational to replace medical care (e.g., inpatient) with prescription drugs even if doing so adversely affects health because there may be costs savings (i.e., reduced copayments associated with medical care), or other

[9]See National Vital Statistics Reports "Deaths: Final Data for 2013" at http://www.cdc.gov/nchs/data/nvsr/nvsr64/nvsr64_02.pdf. Last accessed July 24, 2017.

[10]See for example: Lichtenberg and Sun (2007), Ketcham and Simon (2008), Yin et al. (2008), Schneeweiss et al. (2009), Zhang et al. (2009), and Kaestner and Khan (2012).

[11]Prescription drugs and use of other services may be complementary, for example, if prescription drug use improved health that made treatment for disease more efficacious.

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TABLE 1
Sample Means and Standard Deviations of Hospital Admissions Per 1,000 by Year

	Full Sample					White Only Sample				
Type of Admissions	2002	2004	2006	2008	2009	2002	2004	2006	2008	2009
AMI	8.29	7.25	6.30	5.74	5.36	8.51	7.43	6.44	5.86	5.46
	(3.10)	(2.83)	(2.43)	(2.29)	(2.16)	(3.19)	(2.89)	(2.49)	(2.33)	(2.21)
CHF	10.82	10.54	9.44	8.07	7.75	10.49	10.23	9.12	7.79	7.46
	(5.80)	(6.07)	(5.68)	(5.15)	(4.95)	(5.86)	(6.12)	(5.71)	(5.16)	(4.98)
Coronary atherosclerosis	18.53	17.70	15.83	11.62	9.90	19.04	18.18	16.28	11.92	10.14
	(6.30)	(6.32)	(6.05)	(4.62)	(4.03)	(6.61)	(6.64)	(6.36)	(4.81)	(4.20)
Stroke	6.78	6.07	5.64	5.14	5.02	6.53	5.81	5.43	4.96	4.82
	(2.94)	(2.70)	(2.54)	(2.38)	(2.30)	(2.95)	(2.72)	(2.55)	(2.39)	(2.32)
Cardiac dysrhythmias	11.47	10.83	10.63	10.64	10.38	11.82	11.17	11	11.01	10.77
	(4.53)	(4.52)	(4.50)	(4.56)	(4.53)	(4.62)	(4.63)	(4.60)	(4.62)	(4.63)
Hypertension	1.12	1.05	0.99	1.07	1.10	1.02	0.95	0.89	0.93	0.95
	(0.64)	(0.59)	(0.54)	(0.59)	(0.61)	(0.63)	(0.58)	(0.53)	(0.55)	(0.57)
Angina	1.38	0.93	0.67	0.50	0.43	1.37	0.93	0.66	0.49	0.42
	(0.33)	(0.22)	(0.16)	(0.14)	(0.12)	(0.33)	(0.22)	(0.17)	(0.15)	(0.12)
Pneumonia	9.77	8.98	8.62	7.55	6.98	9.83	9.12	8.77	7.66	7.10
	(4.88)	(4.51)	(4.44)	(4.25)	(3.81)	(4.91)	(4.55)	(4.46)	(4.27)	(3.83)
COPD	7.67	6.44	5.37	6.15	5.79	7.79	6.58	5.47	6.29	5.93
	(2.46)	(2.10)	(1.81)	(2.33)	(2.13)	(2.41)	(2.05)	(1.75)	(2.28)	(2.08)
Dehydration	4.13	3.92	4.07	3.51	2.94	4.08	3.88	4.04	3.5	2.94
	(2.29)	(2.08)	(2.19)	(1.96)	(1.61)	(2.28)	(2.06)	(2.18)	(1.94)	(1.62)
Osteoarthritis	11.88	13.99	14.79	14.77	15.18	12.30	14.55	15.39	15.43	15.87
	(2.90)	(3.62)	(3.64)	(3.71)	(3.74)	(2.80)	(3.53)	(3.53)	(3.62)	(3.62)
UTI	3.29	3.56	3.74	3.98	4.05	3.23	3.52	3.71	3.95	4.02
	(2.09)	(2.39)	(2.65)	(3.02)	(3.13)	(2.08)	(2.40)	(2.67)	(3.03)	(3.15)
Diabetes	2.91	2.73	2.57	2.41	2.28	2.57	2.41	2.26	2.14	2.03
	(0.95)	(0.93)	(0.94)	(0.90)	(0.83)	(0.88)	(0.86)	(0.85)	(0.83)	(0.76)

(*Continued*)

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TABLE 1
Continued

	Full Sample					White Only Sample				
Type of Admissions	2002	2004	2006	2008	2009	2002	2004	2006	2008	2009
Sum 13 conditions	98.06	93.99	88.66	81.15	77.16	98.58	94.76	89.46	81.95	77.92
	(31.90)	(31.06)	(29.75)	(28.82)	(27.22)	(31.97)	(31.10)	(29.70)	(28.62)	(27.10)
Trauma	11.46	11.67	11.70	11.14	10.73	11.83	12.1	12.14	11.59	11.19
	(7.54)	(7.73)	(7.91)	(7.49)	(7.32)	(7.87)	(8.10)	(8.24)	(7.83)	(7.68)
Other admissions	146.81	146.31	145.83	139.32	134.36	147.22	146.76	146.24	139.92	134.96
	(40.02)	(42.28)	(43.42)	(43.22)	(41.38)	(39.94)	(42.13)	(43.10)	(42.71)	(40.89)
All admissions	256.33	251.97	246.19	231.61	222.24	257.63	253.62	247.84	233.47	224.07
	(77.06)	(78.84)	(78.98)	(77.87)	(74.37)	(77.22)	(78.94)	(78.82)	(77.38)	(74.00)
Mortality	32.37	30.22	29.24	28.19	26.87	32.16	30.10	29.12	28.18	26.86
	(19.19)	(18.20)	(17.91)	(17.70)	(16.98)	(19.10)	(18.22)	(17.86)	(17.68)	(16.98)
Observations	115	115	115	115	115	105	105	105	105	105

Notes: The unit of observation is the age-gender-region cell. Figures are weighted by cell size. Standard deviations are in parentheses.

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benefits (e.g., less invasive treatment) associated with using prescription drugs. Health, and therefore hospitalization, may also be unaffected by prescription drug insurance because of moral hazard effects of insurance (health benefits of prescription use that are below the marginal cost of use). Consider that prior to Medicare Part D, data from the Medicare Current Beneficiary Survey (MCBS) indicate that elderly without prescription drug insurance filled a significant number of prescriptions per year (Kaestner and Khan, 2012). Thus, many seniors without prescription drug insurance use at least some of the drugs that may have been essential to maintain health. The additional drug use associated with prescription drug insurance may have modest health benefits and little effect on hospitalizations. In fact, greater use of prescription drugs and use of a greater number of therapeutic classes of drugs may increase adverse drug events, worsen health, and increase hospitalizations (Lazarou, Pomeranz, and Corey, 1998; Tamblyn et al., 2001; Gurwitz et al., 2003; Fu et al., 2010).

In sum, while it is possible that prescription drug insurance may lead to more prescription drug use, improvements in health, and reductions in downstream medical costs such as hospitalizations, it is also possible that the primary effect of prescription drug insurance is to increase the use of drugs with marginal health benefits and little change in hospitalizations or mortality. Indeed, prescription drug insurance may even lead to worse health outcomes and increased hospital and other health care utilization due to adverse drug reactions.

Estimation

Our empirical objective is to estimate the causal effect of gaining prescription drug insurance through Medicare Part D on hospitalization and mortality. The main obstacle to obtaining estimates that can be plausibly interpreted as causal is that elderly with and without prescription drug insurance are likely to differ in ways that affect health and use of services. For example, elderly with prescription drug insurance may be sicker and/or more risk averse than elderly without insurance, and both of these factors would likely affect health and use of other healthcare services. To address this selection problem, we exploit the natural experiment afforded by Medicare Part D, which significantly increased prescription drug insurance coverage.

Using the variation in prescription drug insurance caused by the implementation of Part D, we obtain estimates of the effect of prescription drug insurance on hospitalization, or the treatment-on-treated effect, using an instrumental variables (IV) approach. The following algebraic representation illustrates the IV approach:

$$\mathrm{HOSP}_{it} = \alpha_i + \delta_t + \gamma_{jt} + \beta\ \widehat{\mathrm{INSURED}}_{it} + \lambda \mathrm{MED\Delta D}_{it} + v_{it}, \tag{1}$$

where $i = 1, \ldots, 115$ and $t = 2002, \ldots, 2005, 2007, \ldots, 2009$.

$$\mathrm{INSURED}_{\mathrm{it}} = \tilde{\alpha}_i + \tilde{\delta}_t + \tilde{\gamma}_{jt} + \sum_{t=2002}^{2009} \rho_t(\mathrm{YEAR}_t \times \mathrm{UNINS02\Delta05}_i) + \tilde{\lambda}\mathrm{MED\Delta D}_{it} + u_{it}, \tag{2}$$

where $i = 1, \ldots, 115$ and $t = 2002, \ldots, 2005, 2007, \ldots, 2009$.

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In Equation (1), the hospitalization rate (HOSP) of demographic cell i in year t depends on a cell-specific fixed-effect (a_i), year fixed effects (δ_t), fixed effects for age-by-gender-by-year and region-by-gender-by-year, all denoted by (γ_{jt}) to conserve on notation; the *predicted* proportion of cell i that had prescription drug insurance in period t (INSURED_{it}); and the share of cell i in year t that was in a Medicare Advantage plan (MED_AD).[12] Information on the share of a demographic cell that has insurance comes from the MCBS, and because of small sample sizes per year, we collapse year. Therefore, this variable takes on only two values for each demographic cell—one value in the 2002–2005 period and one value in the 2007–2009 period. However, for 2007 onward, we also have information from Medicare administrative data (MBSF) on the proportion of a demographic cell that is insured in each year from 2007 to 2009. To assess the sensitivity estimates to the source of information on prescription insurance (in post–Part D period), we estimate all models using both sources and report results noting here that results are virtually the same using either source of information.

Equation (2) is the first-stage model that predicts prescription drug insurance. Here, the proportion of cell i in year t with prescription drug insurance depends on demographic cell fixed effects ($\tilde{a}_i$), year fixed effects ($\tilde{\delta}_t$), fixed effects for age-by-gender-by-year, and region-by-gender-by-year ($\tilde{\gamma}_{jt}$), interactions between year dummy variables and the proportion of cell i without prescription drug insurance in 2002–2005 (UNINS_0205_i), and the share of cell i in year t enrolled in Medicare Advantage. We use the symbol $\sim$ to differentiate parameters between Equations (1) and (2). Note that the proportion uninsured in 2002–2005 comes from the MCBS and does not vary by year—it is time invariant. The interactions between the time-invariant rate of uninsured in 2002–2005 in demographic cell i and year dummy variables are the instruments for the rate of prescription drug insurance in Equation (1). In most analyses, we use the specification of the model illustrated by Equations (1) and (2) that uses all years of data. However, because the information on prescription drug insurance taken from the MCBS does not vary by year, we also estimate models using data collapsed to two periods: 2002–2005 and 2007–2009. Results from these analyses are reported below and are virtually identical to those that use all years of data.

Finally, the reduced from model is given by:

$$\text{HOSP}_{it} = \bar{\alpha}_i + \bar{\delta}_t + \bar{\gamma}_{jt} + \sum_{t=2002}^{2009} \bar{\rho}_t (YEAR_t \times \text{UNINS}\Omega 205_i) + \bar{\lambda}\text{MED}\Delta\text{D}_{it} + e_{it}, \qquad (3)$$

where $i = 1, \ldots, 115$ and $t = 2002, \ldots, 2005, 2007, \ldots, 2009$.

[12]Note that the specification includes three of the four of the possible three-way interactions of age, gender, region, and year. A simpler model, for example, cell and year fixed effects, was rejected by tests of the identification assumption (parallel trends). Below we show that the specification underlying Equations (1) and (2) is consistent with identification assumption.

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All variables in Equation (3) have been previously defined, and we use the symbol – to differentiate parameters between Equations (2) and (3). The reduced-form model is helpful for testing the exclusion restriction of the IV approach. We discuss this further below, but note here that the overidentification test is the equivalent of assessing the parallel trend assumption underlying the reduced-form model. Estimates associated with the interactions between the rate of uninsured in 2002–2005 and pre-2006 year dummy variables in Equation (3) should be zero because Part D had not yet been implemented.

Evidence of the Validity of the IV Approach

The identifying assumptions of the IV approach is that the instruments in Equation (2) —the interactions between year dummy variables and the proportion of cell *i* without prescription drug insurance—are sufficiently correlated with prescription drug insurance (i.e., strong first-stage relationship) and are excludable from the outcome equation (Equation (1)). The instruments are strongly correlated with prescription drug insurance. After Medicare Part D, the variation in the proportion of each cell without prescription drug insurance was greatly reduced and cells with a high proportion uninsured in 2002–2005 period experienced the largest decrease in uninsured pre– to post–Medicare Part D. In Table 2, we present the first-stage estimates from Equation (2) and demonstrate that the instruments are strong predictors of prescription drug insurance.

The exclusion restriction is also plausibly valid. The variation in prescription drug insurance pre–Part D is due to differences in insurance by age, gender, and region. It is well known that there are substantial, longstanding regional differences in health insurance by region (Morrisey and Jensen, 1989; Cohen and Makuc, 2008). The variation in health insurance for those under age 65 extends to prescription drug insurance among the elderly, which is not surprising given that a large share of prescription drug insurance is through retiree benefits linked to health insurance coverage. There are also substantial differences in health insurance coverage by gender and age. Again, differences in prescription drug insurance by age and gender among the elderly are expected as access to employee health insurance differ by gender, for example, because of previous work experience and marriage, and age because of declining retiree health insurance coverage and cohort-specific changes in employment (McCormack et al., 2002; McArdle, Neuman, and Huang, 2014).

Given the source of variation in prescription drug insurance, the specification of Equations (1) and (2), which include interactions between the demographic components (i.e., age, region, and gender) that define the cells and year, bolsters the likely validity of the IV approach, as the specification controls for differential trends by age-by-gender and by region-by-gender.

To test the likely validity of the identification assumption, we also estimate a "placebo" analysis to test directly whether trends in hospitalizations and mortality were the same for demographic groups (cells) more or less likely to be without prescription drug insurance prior to Medicare Part D. To do so, we estimate the reduced-form model using only years prior to 2006. We divide the years into two periods: 2002–2003 and 2004–2005 and estimate the following:

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TABLE 2
First-Stage Estimates for Prescription Drug Insurance: Instruments Are Interaction Between Proportion Uninsured Pre–Part D and Year

	Full Sample (White and Nonwhite)		White-Only Sample	
	MCBS	MBSF	MCBS	MBSF
Proportion uninsured by year 2003	0.0004	−0.001	0.001	−0.001
	(0.003)	(0.003)	(0.003)	(0.003)
Proportion uninsured by year 2004	0.002	0.001	0.004	0.0002
	(0.006)	(0.006)	(0.006)	(0.006)
Proportion uninsured by year 2005	0.006	0.003	0.007	0.002
	(0.009)	(0.009)	(0.009)	(0.009)
Proportion uninsured by year 2007	0.760	0.974	0.781	0.982
	(0.107)***	(0.022)***	(0.129)***	(0.023)***
Proportion uninsured by year 2008	0.756	0.967	0.773	0.969
	(0.108)***	(0.026)***	(0.130)***	(0.028)***
Proportion uninsured by year 2009	0.752	0.962	0.767	0.961
	(0.110)***	(0.026)***	(0.132)***	(0.028)***
F-statistics for postperiod interactions	21.11	697.52	19.46	755.91
	(0.0000)	(0.0000)	(0.0000)	(0.0000)
F-statistics for preperiod interactions	0.38	0.61	0.34	0.51
	(0.77)	(0.61)	(0.80)	(0.68)
Cell FE and year FE	Yes	Yes	Yes	Yes
Age-gender-year FE	Yes	Yes	Yes	Yes
Gender-region-year FE	Yes	Yes	Yes	Yes
Observations	805	805	735	735

Notes: MCBS prescription drug insurance information comes from the Medicare Current Beneficiary Survey in both pre– and post–Part D years. MBSF prescription drug insurance information comes from the Medicare Current Beneficiary Survey in pre–Part D years and from the Master Beneficiary Summary File in post–Part D years. Unit of observation is the year-age-gender-region cell. All regressions include HMO share. Regressions are weighted by cell size. Robust, clustered (on cell) standard errors are shown in parentheses. $^{***}p \leq 0.01$.

$$\text{HOSP}_{it} = \bar{\alpha}_i + \bar{\delta}_t + \bar{\gamma}_{jt} + \bar{\rho}(Y0405_t \times \text{UNINS}\Omega205_i) + \bar{\lambda}\text{MED}\Delta\text{D}_{it} + e_{it}, \qquad (4)$$

where $i = 1, \ldots, 115$ and $t = 2002$–2003, 2004–2005.

The key estimate in Equation (4) is the coefficient on the interaction between the baseline rate of uninsured in a cell (UNINSURED_i) and the dummy variable indicating the 2004–2005 period ($Y0405_t$). We report these results in more detail below, but note here that none of the "placebo" estimates were significant (0.05 level). An alternative to the placebo analysis is to estimate the reduced form and test whether estimates associated with the interactions between the rate of uninsured in 2002–2005 and pre-2006 year dummy variables in Equation (3) are zero, which they should be because Part D had not yet been implemented. As expected, these results are consistent with the results of the placebo analysis.

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Second, while the exclusion restriction of the IV analysis cannot be tested in a definitive manner, it is possible to provide evidence of its validity. Note that the IV model is overidentified. As shown in Equation (2), there are six instruments—interactions between year dummy variables and the proportion of cell without prescription drug insurance in 2002–2005. Therefore, we can conduct tests of overidentification, which are analogous to testing whether the pre-2006 interactions between year dummy variables and the proportion of cell without prescription drug insurance in 2002–2005 are excludable. We report these tests below and note here that we cannot reject (0.05 level) the validity of the overidentification restrictions in a large majority of cases. This result combined with the placebo and event-history assessments, provide strong support for the validity of the IV research design.

Interpretation of IV Estimate

The IV approach estimates the effect of prescription drug insurance on hospitalization and mortality for elderly affected by Medicare Part D (local average treatment effect LATE). Kaestner and Khan (2012) examine characteristics of the group likely to be uninsured prior to Part D and report that age, education, income, race, and region are all significant predictors of being uninsured. Overall, those without prescription drug insurance prior to Medicare Part D tend to low educated, low income, black, older, and more likely to come from the east and west south-central divisions. Thus, our estimates of the effect of prescription drug coverage due to Medicare Part D will be applicable to persons with these characteristics, which is a group particularly interesting to public health policy. Our estimates also pertain to the benefit design and generosity of prescription drug insurance as obtained through Medicare Part D.

It is also important to note that the IV approach is not strictly valid if there were general equilibrium effects associated with Medicare Part D, for example, if prices of prescription drugs changed. In addition, there is the possibility that some elderly with prior prescription drug insurance switched insurance plans (i.e., generosity) as a result of Medicare Part D, or the generosity of their insurance plan changed (Levy and Weir, 2010; Engelhadt and Gruber, 2011; Kaestner and Khan, 2012). Given these possible general equilibrium changes, IV estimates measure the effect of gaining coverage among those previously uninsured and the effect of any, arguably minor, change in prices or the generosity of prescription drug insurance among those with prescription drug insurance.[13]

However, changes in prescription drug insurance associated with Medicare Part D are also associated with changes in prescription drug use. Those who gained coverage had a large increase in prescription drug use relative to those who did not gain coverage including those who switch generosity of plans (Zhang et al., 2009; Kaestner and Khan, 2012). So one alternative way to view the IV estimates that is valid even in

[13]Duggan and Morton (2010) conclude that Medicare Part D reduced prices paid to manufacturers of prescription drugs by between 13 percent and 20 percent. To the extent that retail prices of prescription drugs changed post–Medicare Part D, IV estimates include both the effect of prescription drug insurance and effect of price changes. However, since the price effect also affected those who always had insurance, the general price decline does affect both groups.

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light of general equilibrium effects is as the effect of a relative change in prescription insurance coverage that generates approximately a 30–40 percent increase in use of prescription drugs, which is in the middle of the range of estimates in the literature of the change in prescription drug use associated with gaining coverage through Medicare Part D (Lichtenberg and Sun, 2007; Yin et al., 2008; Zhang et al., 2009; Kaestner and Khan, 2012; Dunn and Shapiro, 2015).[14]

RESULTS

First-Stage Estimates

Figure 1 is a scatter plot of the proportion uninsured pre–Part D (i.e., 2002–2005) in a demographic cell against the change in proportion insured pre– (2002–2005) and post– (2007–2009) Part D for the cell. We show a separate figure for the full sample and for the white-only sample. As clearly seen in Figure 1, demographic cells that had lower prescription insurance coverage (higher uninsured rate) before Part D experienced larger gains in insurance coverage pre– to post–Part D. In fact, there is almost a one-for-one (slope of the fitted line is 0.87 for full sample and 0.97 for white sample) gain in insurance coverage for every increase in uninsured rate.

Table 2 presents the key estimates from the first-stage model (Equation (2)) predicting prescription drug insurance coverage. The instruments are the interactions between the proportion of a demographic cell that was uninsured prior to Medicare Part D (2002–2005) and year dummy variables. Implementation of Part D greatly expanded prescription drug insurance coverage and those who gained most were those least likely to be covered prior to 2006. We present estimates for two samples (full and white only) and from models that use either the MCBS information in the post-2006 period or Medicare administrative information (MBSF) on prescription drug coverage after 2006. Standard errors are clustered on cell.

Estimates in Table 2 indicate that for every 1 percentage point increase in the proportion of a cell that was uninsured prior to Part D, prescription drug insurance coverage increased by between 0.75 and 0.98 percentage points. *F*-tests of the significance of the instruments are adequate even with the extensive set of fixed effects included. The *F*-statistics for the model that uses the administrative data are significantly larger because that data have separate values of insurance coverage for each cell-year combination post-2006—the MCBS has only one value for each demographic cell in the post-2006 period. Overall, Figure 1 and estimates in Table 2 show that the instruments are strong predictors of prescription drug insurance and that Medicare Part D significantly increased insurance, particularly among groups not well insured prior to Part D.

Hospitalizations

Table 3 presents ordinary least squares (OLS) and IV estimates of the effect of gaining prescription drug insurance on hospitalizations for two samples: the full sample and a sample restricted to white persons. For each sample and each type of hospitalization,

[14]For studies that estimated intention-to-treat effects of Medicare Part D on prescription drug use, we scale the estimate by the change in prescription drug insurance.

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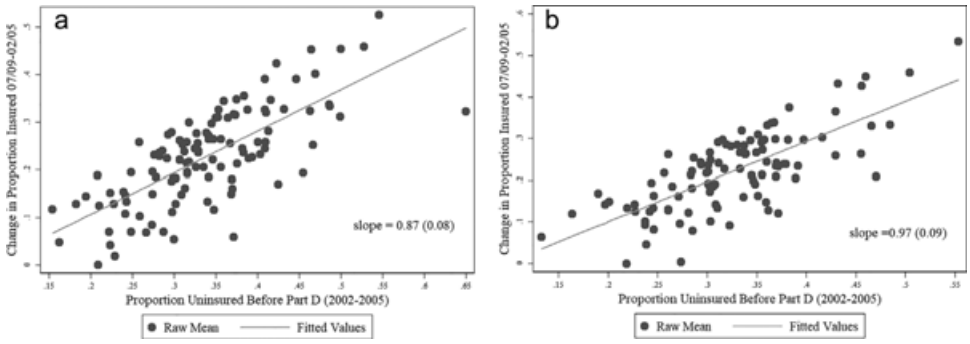
FIGURE 1
Association Between Change in Insurance Coverage Pre– to Post–Part D and Proportion Uninsured Pre–Part D: (a) Full Sample (White and Nonwhite). (b) White Only



Notes: Information on prescription insurance comes from Medicare Current Beneficiary Survey for both 2002–2005 (pre–Part D) and 2007–2009 (post–Part D). Raw mean is the proportion uninsured (insured) for an age-gender-region demographic cell. Fitted values are obtained from a regression of change in proportion insured on proportion of uninsured pre–Part D without covariates

we report three estimates: OLS obtained using MCBS to measure prescription insurance coverage pre– and post–Part D; IV estimates obtained using MCBS to measure prescription insurance coverage pre– and post–Part D; and IV estimates obtained using MCBS to measure prescription insurance coverage pre–Part D and administrative data (MBSF) to measure prescription drug insurance post–Part D.

The first point of note about estimates in Table 3 is that 75 percent of the time IV estimates are more negative than OLS estimates. This is exactly what would be expected if there is selection into prescription drug coverage that is driven by poor health, which is a main prediction of economic theory. The second point to note is that IV estimates obtained using the two different sources of information about prescription drug insurance are very similar. The main difference is that the IV estimates obtained using the administrative information (MBSF) on prescription drug insurance are more precisely estimated, which is consistent with the stronger first-stage correlation. Given the similarity of estimates and the greater precision of the IV estimates obtained using the MBSF, we focus on these estimates in our discussion.

IV (MBSF) estimates in Table 3 indicate that gaining Medicare Part D prescription drug coverage is associated with a significant decrease in CHF and COPD admissions for both the full sample and white-only sample. For the full sample, gaining prescription drug coverage through Medicare Part D is associated with a 1.4 per 1,000 (13 percent) decrease in CHF admissions and a 1.3 per 1,000 (19 percent) decrease in COPD admissions. Analogous figures for the white-only sample are 2.5 per 1,000 (25 percent) and 1.9 per 1,000 (28 percent), respectively. For the white sample, there is evidence that gaining prescription drug insurance was

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TABLE 3
Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on Hospital Admissions

	Full Sample (White and Nonwhite)			White-Only Sample		
Type of Hospital Admission	OLS	IV MCBS	IV MBSF	OLS	IV MCBS	IV MBSF
Acute myocardial infarction (AMI)	0.05	−0.61	−0.47	0.18	−0.39	−0.31
	(0.48)	(0.45)	(0.34)	(0.56)	(0.48)	(0.38)
Weighted mean 2002–2005 = 7.50 (F), 7.68 (W)						
Congestive heart failure (CHF)	−0.45	−1.79**	−1.39**	−0.75	−3.19***	−2.53***
	(0.61)	(0.78)	(0.59)	(0.76)	(0.93)	(0.63)
Weighted mean 2002–2005 = 10.51 (F), 10.18 (W)						
Coronary atherosclerosis	−0.33	0.47	0.36	−0.62	−0.87	−0.69
	(0.64)	(0.90)	(0.69)	(0.81)	(0.89)	(0.71)
Weighted mean 2002–2005 = 17.56 (F), 18.04 (W)						
Stroke	0.18	0.09	0.07	0.18	0.07	0.06
	(0.36)	(0.42)	(0.32)	(0.41)	(0.50)	(0.39)
Weighted mean 2002–2005 = 6.24 (F), 6.00 (W)						
Cardiac dysrhythmias	−0.62	0.03	0.02	−0.69	−0.52	−0.42
	(0.49)	(0.57)	(0.44)	(0.60)	(0.60)	(0.48)
Weighted mean 2002–2005 = 10.93 (F), 11.28 (W)						
Hypertension	0.24	0.20	0.15	0.12	−0.01	−0.01
	(0.14)	(0.21)	(0.16)	(0.13)	(0.20)	(0.16)
Weighted mean 2002–2005 = 1.05 (F), 0.95 (W)						
Angina	−0.05	−0.12	−0.09	−0.09	−0.26	−0.20
	(0.10)	(0.12)	(0.09)	(0.12)	(0.13)	(0.11)
Weighted mean 2002–2005 = 1.04 (F), 1.03 (W)						
Sum: Heart/circulatory diseases	−0.99	−1.73	−1.35	−1.66	−5.17***	−4.11***
	(1.39)	(1.51)	(1.19)	(1.69)	(1.67)	(1.24)
Weighted mean 2002–2005 = 54.84 (F), 55.16 (W)						
Pneumonia	0.15	0.42	0.32	0.03	−0.34	−0.27
	(0.49)	(0.54)	(0.42)	(0.49)	(0.56)	(0.44)
Weighted mean 2002-05 = 9.60 (F), 9.72 (W)						
COPD	−1.06	−1.67**	−1.30**	−1.74**	−2.42***	−1.92***
	(0.56)	(0.67)	(0.54)	(0.67)	(0.69)	(0.58)
Weighted mean 2002–2005 = 6.85 (F), 6.98 (W)						
Dehydration	−0.15	−0.60	−0.47	−0.39	−0.75**	−0.60**
	(0.33)	(0.36)	(0.28)	(0.33)	(0.35)	(0.27)
Weighted mean 2002–2005 = 3.98 (F), 3.94 (W)						
Osteoarthritis	0.49	−0.37	−0.28	0.59	−0.48	−0.38
	(0.62)	(0.80)	(0.62)	(0.72)	(0.96)	(0.75)

(*Continued*)

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TABLE 3
Continued

	Full Sample (White and Nonwhite)			White-Only Sample		
Type of Hospital Admission	OLS	IV MCBS	IV MBSF	OLS	IV MCBS	IV MBSF
Weighted mean 2002–2005 = 13.35 (F), 13.86 (W)						
Urinary tract infection (UTI)	0.49	0.59	0.46	0.16	0.50	0.40
	(0.45)	(0.63)	(0.49)	(0.49)	(0.64)	(0.50)
Weighted mean 2002–2005 = 3.49 (F), 3.44 (W)						
Diabetes	−0.04	−0.68**	−0.53**	−0.27	−0.57	−0.45
	(0.25)	(0.31)	(0.23)	(0.31)	(0.34)	(0.27)
Weighted mean 2002–2005 = 2.76 (F), 2.43 (W)						
Sum of the 13 conditions	−1.11	−4.04	−3.15	−3.27	−9.23***	−7.33***
	(2.20)	(2.43)	(1.90)	(2.65)	(2.55)	(1.86)
Weighted mean 2002–2005 = 94.87 (F), 95.54 (W)						
Trauma	0.34	0.94	0.73	0.17	0.59	0.47
	(0.80)	(1.10)	(0.84)	(0.96)	(1.12)	(0.89)
Weighted mean 2002–2005 = 11.59 (F), 12.00 (W)						
Other admissions	−2.14	−9.82**	−7.66***	−1.50	−8.83**	−7.03**
	(3.10)	(3.90)	(2.82)	(3.70)	(3.87)	(3.09)
Weighted mean 2002–2005 = 146.07 (F), 146.52 (W)						
All admissions	−2.91	−12.93**	−10.08**	−4.61	−17.47***	−13.89***
	(4.82)	(5.46)	(4.11)	(6.09)	(6.01)	(4.66)
Weighted mean 2002–2005 = 252.53 (F), 254.06 (W)						
Cell FE and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Age-gender-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Gender-region-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	805	805	805	735	735	735

Notes: MCBS prescription drug insurance information comes from the Medicare Current Beneficiary Survey in both pre– and post–Part D years. MBSF prescription drug insurance information comes from the Medicare Current Beneficiary Survey in pre–Part D years and from the Master Beneficiary Summary File in post–Part D years. Unit of observation is the year (period)-age-gender-region cell. The year 2006 is excluded. All regressions include HMO share. Regressions are weighted by cell size. Robust, clustered (on cell) standard errors are shown in parentheses. $^{**}p \leq 0.05$, $^{***}p \leq 0.01$.

associated with a 4.1 per 1,000 (7 percent) decrease in all heart/circulatory admissions combined. Estimates in Table 3 also indicate declines in diabetes associated with gaining prescription drug coverage, although only estimates for the full sample are statistically significant (0.05 level). For both samples, gaining coverage is associated with approximately a 0.5 per 1,000 (19 percent) decrease in diabetes admissions. Among whites, gaining prescription drug insurance is also associated with a 0.6 per 1,000 (15 percent) decline in admissions for dehydration.

Given that there are 13 different outcomes, there may be some concern from multiple testing bias. However, we note that we also show summary measures in the second

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part of Table 3 to address the possible bias from multiple testing. Nevertheless, we apply a Holm–Bonferroni correction and use this more conservative, perhaps overly conservative, standard to assess statistical inference.[15] Online Appendix Table 1 (Kaestner, Schiman and Alexander, 2017) provides the comparisons. Applying the more conservative standard suggests that only the CHF and COPD estimates for whites remain statistically significant.

The second part of Table 3 presents estimates for aggregated groups of admissions: sum of the 13 individual categories, trauma, and other admissions. These 3 categories are mutually exclusive and sum to all admissions, which are also shown. For both samples, gaining prescription drug coverage is associated with statistically significant declines in admissions other than the 13 listed in Table 3, and in all admissions. The effect sizes are approximately 5 percent for both samples (4 percent for all admissions in the full sample). For whites, estimates also indicate a statistically significant decline in admissions for the 13 conditions combined; gaining coverage is associated with a 7.3 per 1,000 (8 percent) decline in these admissions. Finally, estimates pertaining to trauma are positive and not statistically significant. We expected no effect of prescription drug insurance on trauma, which is mainly injuries and other conditions unlikely to respond to greater prescription drug use.

Online Appendix Table 2 (Kaestner, Schiman and Alexander, 2017) presents IV estimates analogous to Table 3, but from models that do not weight by cell size. Estimates in this table are very similar to those in Table 3, so weighting does not matter. Online Appendix Table 3 (Kaestner, Schiman and Alexander, 2017) presents estimates analogous to Table 3, but from models that use data collapsed to two periods: 2002–2005 and 2007–2009. Estimates from these analyses are virtually the same as those reported in Table 3.

To assess the likely validity of the IV estimates reported in Table 3, we conduct three analyses: a placebo test using only pre–Part D data; event-study, reduced-form analyses; and tests of overidentification. These tests are conceptually related and all test the plausibility of the exclusion restriction, which is that the instruments—the interactions between baseline (2002–2005) rates of uninsured and year—are excludable from the outcome equation. The exclusion restriction is the parallel trend assumption of a difference-in-differences approach because the first stage of the IV method is a difference-in-differences approach.

Online Appendix Table 4 (Kaestner, Schiman and Alexander, 2017) reports the estimates from the placebo analysis. None of the 36 estimates are statistically significant. Online Appendix Table 5 (Kaestner, Schiman and Alexander, 2017) reports the p-values from the overidentification tests associated with each IV estimate in Table 3. We cannot reject (0.05 level) the validity of the exclusion restrictions in 52 out of 72 cases (72 percent of the time). Given that the IV approach is based on the same assumption for each outcome, the failure to reject the exclusion restrictions in most cases provides considerable support for the validity of the research design.

[15]The Holm–Bonferroni correction is overly conservative if tests are not independent, which they are unlikely to be.

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Finally, in Online Appendix Table 6 (Kaestner, Schiman and Alexander, 2017), we report estimates from the reduced-form model corresponding to Table 3. The key aspect of these estimates are the *F*-tests of the joint hypothesis that the interactions between the rate of uninsured in 2002–2005 and pre-2006 year dummy variables are zero. These tests are similar to the overidentification tests reported earlier. As indicated in the table, none of the *F*-tests can reject the hypothesis of equivalent time trends in hospitalization prior to implementation of Part D for demographic cells more or less likely to be uninsured prior to Part D. It is also the case, however, that the post-2006 interactions between the rate of uninsured in 2002–2005 and year dummy variables are not significant in most cases. However, a careful review of estimates suggests results that are consistent with the IV results reported in Table 3. To make this easier to see, we reestimate the reduced-form model using data collapsed to two periods. These estimates are reported in Online Appendix Table 3 (Kaestner, Schiman and Alexander, 2017) and are very similar to the IV estimates in Table 3.

Medicare Hospital Payments

In Table 4, we present OLS and IV estimates of the effect of gaining prescription drug coverage for total Medicare payments. The organization of Table 4 is the same as that used for Table 3. The IV (MBSF) estimates in Table 4 are generally consistent with those in Table 3—significant decreases in the rate of admission per 1,000 persons are associated with decreases in Medicare payments, although estimates in Table 4 are not always statistically significant even though they are often of the same magnitude of those in Table 3. For the full sample, estimates in Table 3 indicate that gaining prescription drug coverage is associated with a 13 percent decrease in CHF admissions and 19 percent decrease in COPD admissions. The corresponding estimates in Table 4 indicate that gaining prescription drug coverage is associated with a 10 percent decrease in Medicare payments for CHF admissions and a 21 percent decrease in Medicare payments for COPD admissions. For the white-only sample, estimates in Table 3 indicate a 25 percent and 28 percent decrease in CHF admissions and COPD admissions, respectively. Corresponding estimates in Table 4 indicate a 23 percent and 28 percent decrease in Medicare payments for these admissions. Similarly, estimates in Table 3 indicate that gaining prescription drug coverage is associated with approximately a 5 percent decrease in admissions other than the 13 conditions and for all conditions; estimates in Table 4 for the same outcomes are of similar magnitude. Overall, although less precisely estimated, estimates of the effect of gaining prescription drug coverage on Medicare payments suggest a decline in payments for those illnesses that also experienced a significant decline in admissions and for all admissions. The declines in payment are of approximately the same magnitude as the decline in admissions.

Tests of the IV approach pertaining to Medicare payments are in Online Appendix Tables 7 and 8 (Kaestner, Schiman and Alexander, 2017). None of the placebo estimates (Online Appendix Table 7) are statistically significant. For the overidentification tests (Online Appendix Table 8), we cannot reject the exclusion restrictions 88 percent of the time (64 out of 72).

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TABLE 4
Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on Medicare Inpatient Payment

	Full Sample (White and Nonwhite)			White-Only Sample		
Type of Hospital Admission	OLS	IV MCBS	IV MBSF	OLS	IV MCBS	IV MBSF
Acute myocardial infarction (AMI)	−4.1	−5.8	−4.5	−3.9	−5.2	−4.1
	(7.3)	(7.7)	(6.0)	(8.9)	(8.6)	(6.9)
Weighted mean 2002–2005 = 96.7 (F), 98.9 (W)						
Congestive heart failure (CHF)	−1.4	−7.0	−5.5	−3.3	−14.7***	−11.7***
	(4.1)	(4.4)	(3.5)	(4.8)	(5.0)	(3.8)
Weighted mean 2002–2005 = 54.4 (F), 51.9 (W)						
Coronary atherosclerosis	−2.0	16.7	13.0	−2.3	3.0	2.4
	(9.4)	(13.6)	(9.9)	(12.8)	(13.7)	(10.8)
Weighted mean 2002–2005 = 207.8 (F), 214.4 (W)						
Stroke	−3.3	2.3	1.8	−5.5	−5.0	−3.9
	(5.0)	(6.1)	(4.7)	(6.0)	(7.7)	(6.1)
Weighted mean 2002–2005 = 47.0 (F), 44.4 (W)						
Cardiac dysrhythmias	−4.8	6.1	4.8	−4.1	3.4	2.6
	(6.5)	(7.0)	(5.4)	(7.5)	(7.5)	(5.9)
Weighted mean 2002–2005 = 76.1 (F), 78.4 (W)						
Hypertension	0.8	1.2	1.0	0.5	1.0	0.8
	(0.7)	(1.0)	(0.8)	(0.7)	(1.0)	(0.8)
Weighted mean 2002–2005 = 3.2 (F), 2.8 (W)						
Angina	−0.5	−0.8	−0.6	−0.5	−1.1**	−0.9**
	(0.3)	(0.4)	(0.3)	(0.4)	(0.5)	(0.4)
Weighted mean 2002–2005 = 2.8 (F), 2.7 (W)						
Sum: heart/circulatory diseases	−15.4	12.8	10.0	−19.0	−18.6	−14.7
	(17.8)	(19.9)	(15.1)	(21.7)	(21.8)	(17.7)
Weighted mean 2002–2005 = 488.0 (F), 493.4 (W)						
Pneumonia	−1.1	1.7	1.3	−1.4	−2.9	−2.3
	(3.9)	(3.8)	(2.9)	(4.0)	(4.5)	(3.5)
Weighted mean 2002–2005 = 49.2 (F), 49.3 (W)						
COPD	−5.7	−8.4**	−6.6	−8.4**	−11.3***	−9.0***
	(3.4)	(4.2)	(3.4)	(4.0)	(4.0)	(3.4)
Weighted mean 2002–2005 = 31.6 (F), 31.8 (W)						
Dehydration	−1.2	−2.7	−2.1	−1.9	−2.7	−2.1
	(1.6)	(1.9)	(1.5)	(1.6)	(1.8)	(1.4)
Weighted mean 2002–2005 = 16.5 (F), 16.1 (W)						
Osteoarthritis	3.7	−1.4	−1.1	4.8	−5.0	−3.9
	(6.8)	(9.7)	(7.6)	(7.8)	(12.6)	(9.8)
Weighted mean 2002–2005 = 127.0 (F), 131.7 (W)						

(*Continued*)

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TABLE 4
Continued

Type of Hospital Admission	Full Sample (White and Nonwhite) OLS	IV MCBS	IV MBSF	White-Only Sample OLS	IV MCBS	IV MBSF
Urinary tract Infection (UTI)	3.9	5.4**	4.2**	3.1	4.7	3.8
	(2.0)	(2.6)	(2.0)	(2.2)	(2.6)	(2.0)
Weighted mean 2002–2005 = 13.6 (F), 13.3 (W)						
Diabetes	−1.1	−3.0	−2.3	−2.8	−3.3	−2.7
	(2.1)	(2.8)	(2.2)	(2.7)	(3.2)	(2.6)
Weighted mean 2002–2005 = 17.3 (F), 15.1 (W)						
Sum of the 13 conditions	−16.9	4.4	3.5	−25.6	−39.1	−31.0
	(22.5)	(26.9)	(20.8)	(28.8)	(30.2)	(24.1)
Weighted mean 2002–2005 = 743.2 (F), 750.7 (W)						
Trauma	−2.3	−7.1	−5.5	−5.7	−21.7	−17.2
	(8.0)	(12.4)	(9.7)	(9.0)	(13.2)	(9.8)
Weighted mean 2002–2005 = 84.0 (F), 86.5 (W)						
Other admissions	−19.9	−57.5	−44.8	3.2	−55.0	−43.7
	(40.8)	(41.6)	(32.7)	(51.9)	(49.1)	(39.1)
Weighted mean 2002–2005 = 1,126.7 (F), 1,128.4 (W)						
All admissions	−39.1	−60.1	−46.8	−28.2	−115.8	−91.9
	(57.3)	(61.2)	(48.7)	(75.7)	(72.3)	(57.2)
Weighted mean 2002–2005 = 1,953.9 (F), 1,965.7 (W)						
Cell FE and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Age-gender-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Gender-region-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	805	805	805	735	735	735

Notes: MCBS prescription drug insurance information comes from the Medicare Current Beneficiary Survey in both pre– and post–Part D years. MBSF prescription drug insurance information comes from the Medicare Current Beneficiary Survey in pre–Part D years and from the Master Beneficiary Summary File in post–Part D years. Unit of observation is the year (period)-age-gender-region cell. The year 2006 is excluded. All regressions include HMO share. Regressions are weighted by cell size. Robust, clustered (on cell) standard errors are shown in parentheses. $^{**}p \leq 0.05$, $^{***}p \leq 0.01$.

Hospital Charges

As discussed, there is relatively strong consistency between estimates in Tables 3 and 4, particularly in terms of magnitudes of effects. This is not surprising given the prospective payment structure based on the DRG classification system. Declines in payment should largely parallel declines in admissions because every admission in a DRG is reimbursed the same. However, prescription drug coverage and the greater use of prescription drugs it engenders may also affect the severity of an admission in addition to the probability of an admission. Hospital charges, which include all resource use for an admission will reflect both the decline in admissions and a change in severity.[16]

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TABLE 5
Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on Medicare Inpatient Charges

	Full Sample (White and Nonwhite)			White-Only Sample		
Type of Hospital Admission	OLS	IV MCBS	IV MBSF	OLS	IV MCBS	IV MBSF
Acute myocardial infarction (AMI)	6.7	−10.0	−7.8	4.1	−6.1	−4.8
	(25.7)	(30.2)	(23.4)	(31.5)	(29.9)	(23.9)
Weighted mean 2002–2005 = 317.0 (F), 324.4 (W)						
Congestive heart failure (CHF)	−2.5	−13.2	−10.3	−7.2	−43.8**	−34.8**
	(16.4)	(19.6)	(15.3)	(18.2)	(19.2)	(14.8)
Weighted mean 2002–2005 = 180.7 (F), 174.1 (W)						
Coronary atherosclerosis	−18.5	16.2	12.6	−20.5	−15.9	−12.7
	(30.5)	(41.7)	(31.9)	(40.9)	(44.8)	(35.9)
Weighted mean 2002–2005 = 641.9 (F), 662.0 (W)						
Stroke	−11.2	7.7	6.1	−10.2	−21.9	−17.2
	(17.4)	(17.9)	(14.0)	(18.3)	(24.3)	(18.8)
Weighted mean 2002–2005 = 141.7 (F), 134.1 (W)						
Cardiac dysrhythmias	−12.8	17.2	13.3	−11.7	−0.7	−0.7
	(24.9)	(26.9)	(20.8)	(30.0)	(29.4)	(23.3)
Weighted mean 2002–2005 = 248.3 (F), 255.9 (W)						
Hypertension	−0.5	−0.6	−0.5	−0.4	−1.2	−0.9
	(3.0)	(3.9)	(3.1)	(3.6)	(4.6)	(3.6)
Weighted mean 2002–2005 = 11.8 (F), 10.4 (W)						
Angina	−3.0**	−3.0	−2.4	−2.6	−2.7	−2.1
	(1.4)	(1.9)	(1.5)	(1.6)	(1.9)	(1.5)
Weighted mean 2002–2005 = 9.7 (F), 9.4 (W)						
Sum: heart/circulatory diseases	−41.7	14.1	11.0	−48.5	−92.2	−73.3
	(61.4)	(65.5)	(50.8)	(73.8)	(73.9)	(59.0)
Weighted mean 2002–2005 = 1551.0 (F), 1570.5 (W)						
Pneumonia	−11.8	−14.0	−11.1	−12.5	−44.0	−35.2
	(15.9)	(19.3)	(15.0)	(17.1)	(25.1)	(18.2)
Weighted mean 2002–2005 = 161.1 (F), 162.6 (W)						
COPD	−27.4**	−48.8**	−38.0***	−40.9***	−59.5***	−47.2***
	(13.3)	(15.6)	(12.5)	(13.4)	(15.0)	(12.2)
Weighted mean 2002–2005 = 103.8 (F), 105.5 (W)						
Dehydration	−8.5	−15.0**	−11.7**	−8.9	−13.1	−10.4
	(6.0)	(6.9)	(5.5)	(6.1)	(7.5)	(6.0)
Weighted mean 2002–2005 = 51.1 (F), 50.2 (W)						
Osteoarthritis	11.1	−87.5	−68.1	8.1	−101.4	−80.4
	(29.5)	(58.0)	(42.7)	(34.5)	(78.7)	(57.6)
Weighted mean 2002–2005 = 402.3 (F), 417.7 (W)						
Urinary tract infection (UTI)	17.4	16.8	13.1	16.2	10.9	8.6
	(9.5)	(11.0)	(8.6)	(9.9)	(12.2)	(9.8)
Weighted mean 2002–2005 = 45.9 (F), 44.8 (W)						
Diabetes	−1.8	−3.7	−2.9	−10.8	−9.2	−7.3
	(9.3)	(10.8)	(8.4)	(10.8)	(12.8)	(10.2)

(*Continued*)

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TABLE 5
Continued

Type of Hospital Admission	Full Sample (White and Nonwhite) OLS	IV MCBS	IV MBSF	White-Only Sample OLS	IV MCBS	IV MBSF
Weighted mean 2002–2005 = 54.2 (F), 47.6 (W)						
Sum of the 13 conditions	−62.6	−138.1	−107.7	−97.4	−308.6**	−245.3**
	(79.6)	(106.8)	(82.8)	(96.1)	(142.1)	(103.2)
Weighted mean 2002–2005 = 2,369.4 (F), 2,398.8 (W)						
Trauma	−9.0	−73.0	−56.8	−6.8	−152.0**	−120.7**
	(30.5)	(54.5)	(42.2)	(33.5)	(68.4)	(46.9)
Weighted mean 2002–2005 = 272.4 (F), 281.6 (W)						
Other admissions	−73.2	−584.9**	−456.0**	−3.0	−700.8**	−557.1**
	(180.5)	(242.3)	(179.0)	(248.3)	(312.5)	(221.0)
Weighted mean 2002–2005 = 3,359.2 (F), 3,366.5 (W)						
All admissions	−144.8	−796.0**	−620.6**	−107.2	−1161.4**	−923.0***
	(248.7)	(360.4)	(270.8)	(339.0)	(488.3)	(340.0)
Weighted mean 2002–2005 = 6,001.0 (F), 6,046.9 (W)						
Cell FE and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Age-gender-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Gender-region-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	805	805	805	735	735	735

Notes: MCBS prescription drug insurance information comes from the Medicare Current Beneficiary Survey in both pre– and post–Part D years. MBSF prescription drug insurance information comes from the Medicare Current Beneficiary Survey in pre–Part D years and from the Master Beneficiary Summary File in post–Part D years. Unit of observation is the year (period)-age-gender-region cell. The year 2006 is excluded. All regressions include HMO share. Regressions are weighted by cell size. Robust, clustered (on cell) standard errors are shown in parentheses. $^{**}p \leq 0.05$, $^{***}p \leq 0.01$.

Estimates of the effect of prescription drug coverage on hospital charges are presented in Table 5, which has the same format as previous tables. Estimates in Table 5 align very well with estimates in Table 3, which is expected, as declines in admissions cause a related decline in hospital charges. This is similar to the correspondence between estimates in Tables 3 and 4. What is different about estimates in Table 5 is that for COPD, other admissions (besides 13 individual types listed), and all admissions, the effect of gaining coverage on hospital charges is substantially above the effect of gaining coverage on admissions. Consider COPD. Estimates in Table 3 indicate that prescription drug coverage is associated with a 19 percent decrease in COPD admissions for the full sample and a 28 percent decrease in COPD admissions for the white-only sample. Corresponding estimates in Table 5 are 37 percent and 45 percent, respectively. Estimates in Table 3 indicate about a 5 percent decline in admissions other than the 13 listed and in all admissions for both the full and white-only samples. In Table 5, estimates suggest that gaining prescription drug coverage is associated with between a 10 percent and 17 percent decrease in hospital charges for these broad groups of admissions. The larger effects of prescription drug coverage on hospital charges

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suggests that prescription drug insurance reduced resource-intensive admissions (i.e., relatively severe cases) more than other admissions.

We also test the identification assumption for this outcome and results are in Online Appendix Tables 9 and 10 (Kaestner, Schiman and Alexander, 2017). Only 1 of the 36 placebo estimates (Online Appendix Table 9) is statistically significant. For the overidentification tests (Online Appendix Table 10), we cannot reject the exclusion restrictions 77 percent of the time (56 out of 72).

Mortality

In Table 6, we present OLS and IV estimates of the effect of prescription drug insurance on all-cause mortality. The presentation follows that of previous tables. Estimates indicate that gaining prescription drug insurance through Medicare Part D is not associated with a significant or clinically meaningful change in mortality for the full sample. The estimate is 0.15 per thousand, which is less than 0.5 percent. For the white-only sample, gaining prescription drug coverage is associated with a 1.5 percent decrease in mortality. Standard errors of estimates in Table 6 suggest that we are able to detect reliably an effect size of approximately 5 percent, so we cannot rule out with statistical certainty a small decrease in mortality.

CONCLUSIONS

Evidence from clinical trials establishes that prescription drugs improve health. Indeed, for some drugs such as statins, antihypertensive drugs, and blood thinners, the (cost) effectiveness of pharmaceutical therapy is substantial (Musini et al., 2009; Taylor et al., 2013). There is also ample evidence that prescription drug insurance, including Medicare Part D, significantly increases prescription drug use (see, e.g.,

TABLE 6
Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on All-Cause Mortality

	Full Sample (White and Nonwhite)			White-Only Sample		
	OLS	IV MCBS	IV MBSF	OLS	IV MCBS	IV MBSF
Mortality	0.11	0.19	0.15	−0.22	−0.60	−0.47
	(0.78)	(1.01)	(0.78)	(0.85)	(1.06)	(0.84)
Weighted mean 2002–2005 = 30.67 (F), 30.51 (W)						
Cell FE and year FE	Yes	Yes	Yes	Yes	Yes	Yes
Age-gender-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Gender-region-year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	805	805	805	735	735	735

Notes: MCBS prescription drug insurance information comes from the Medicare Current Beneficiary Survey in both pre– and post–Part D years. MBSF prescription drug insurance information comes from the Medicare Current Beneficiary Survey in pre–Part D years and from the Master Beneficiary Summary File in post–Part D years. Unit of observation is the year (period)-age-gender-region cell. The year 2006 is excluded. All regressions include HMO share. Regressions are weighted by cell size. Robust, clustered (on cell) standard errors are shown in parentheses.

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Goldman , Joyce, and Zheng, 2007; Yin et al., 2008; Zhang et al., 2009; Kaestner and Khan, 2012; Dunn and Shapiro, 2015). Combined, these two facts imply that gaining prescription drug insurance through Medicare Part D improves health and reduces the use of, and spending on, nonpharmacy medical services. However, there are reasons to expect that this logic may not be correct and that gaining prescription drug insurance through Medicare Part D does not affect the use of other medical services. Most importantly, there may be a relatively small health benefits from greater use of prescription drugs that comes with insurance such as Medicare Part D because of moral hazard. Prior to gaining prescription drug insurance, many elderly bought a substantial number of prescription drugs, and if so, it is likely that they bought the most effective drugs in terms of health benefits.

To date, the evidence on whether prescription drug insurance improves health (mortality) and results in offsets—savings from reduced spending on other services—is inconclusive. To our knowledge there are only three studies that examine whether there are offsets associated with gaining prescription insurance through Medicare Part D. Zhang et al. (2009), using a sample of elderly in one health plan from Pennsylvania, find that a 74 percent increase in prescription drug spending associated with Medicare Part D results in a 7 percent decrease in spending on nonpharmacy medical care. Afendulis et al. (2011), using aggregate state-level data, report that the 28 percentage point increase in prescription drug insurance through Medicare Part D is associated with a 4 percent decrease in hospitalizations. Finally, Kaestner and Khan (2012), using a sample from the MCBS, report that gaining prescription drug insurance through Medicare Part D is not significantly associated with changes in inpatient or outpatient spending, although the study lacks statistical power to detect small to moderate effects. Other evidence related to the question, but not specifically in the context of Medicare Part D, is mixed, as described in the literature review earlier.

Given the importance of the issue for both theory and policy, and the lack of a *consensus* answer, we examine whether gaining prescription drug insurance through Medicare Part D is associated with changes in rates of, and expenditures on, several types of hospitalizations and all hospitalizations combined for a large, geographically diverse group of Medicare beneficiaries. We also study the effect of Part D on all-cause mortality. Our estimates indicate that gaining prescription drug insurance through Medicare Part D is associated with approximately a 4 percent decrease in the number of hospital admissions per 1,000 people, a 2 percent (full sample) to 5 percent (white-only sample) decrease in Medicare payments per person for hospital admissions, and a 10 percent (full sample) to 15 percent (white-only sample) decrease in inpatient charges per person. These estimates are smaller than reported in the most closely related article. Afendulis et al. (2011) report a 4.4 percent decline in admissions associated with a 28 percentage point change in prescription drug insurance coverage. If this estimate is extrapolated to a 100 percentage point change, which is the equivalent of the change in insurance coverage we exploit, the implied effect is 16 percent and nearly three times larger than our estimate.

We find particularly large effects of prescription drug insurance on hospital admissions for CHF, COPD, and diabetes, which are illnesses for which there are effective pharmaceutical treatments. For example, according to a Cochrane review,

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pharmaceutical treatment of COPD results in a 25 percent reduction in exacerbations that cause hospitalization (Nannini et al., 2013). We find that prescription coverage decreases COPD admissions by 19 percent for the full sample and 28 percent for the white-only sample. While not an apples-to-apples comparison, the effect sizes are plausible vis-à-vis that reported in the Cochrane review. Gaining prescription drug coverage through Part D is associated with a 30–40 percent increase in prescription drug use. While this is not the same as being treated or not treated with COPD medications, as in the Cochrane review, it does represent a substantial intervention and increase in medication. In short, based on clinical evidence, our estimate for COPD, which is the largest, is reasonable. Similarly, evidence for some heart disease treatments suggest large effects. A Cochrane review (Taylor et al., 2013) reports that statin therapy is associated with a 32 percent decrease in fatal and nonfatal cardiovascular disease events. We find that prescription drug coverage is associated with a 2 percent decline in heart/circulatory admissions for the full sample and 7 percent decline in such admissions for white persons. Here, too, we are not making an apples-to-apples comparison, but the results from clinical trials are not inconsistent with our results.

We also find effects for "other" admissions. While the clinical linkages between greater prescription drug use and these "other" conditions are not as well known, as compared to certain illnesses such as cardiovascular disease, there is evidence that Part D increases use of all therapeutic classes of drugs (Khan and Kaestner, 2012) and that the "price" elasticity of demand for prescription drugs are relatively similar for chronic and nonchronic treatments (Einav, Finkelstein, and Polyakova, 2016). So, it is plausible that hospitalizations for "other" illnesses declined, consistent with our results.

Our estimates also indicate that Medicare Part D prescription insurance is associated with changes in the severity of admissions as the change in hospital charges associated with prescription drug coverage was usually greater than the change in the quantity of admissions associated with prescription drug coverage, although the charge data may not accurately reflect resource use as we assume.

We find no evidence that gaining prescription drug insurance through Medicare Part D affects all-cause mortality, although we cannot rule out a small decrease of 5 percent or less. This is consistent with the small effects reported in Dunn and Shapiro (2015), but smaller than those reported in Huh and Reif. In fact, the confidence interval around our estimate rules out the 10 percent effect size reported by Huh and Reif (2017). Huh and Reif focus on those 66 years old whereas we analyze a population 66–84 years of age, and we use different methods, most notably not relying on near-elderly as a comparison group.

Overall, Medicare Part D prescription insurance is associated with a $47 (full sample) to $92 (white-only sample) decrease in Medicare inpatient expenditures per person. This figure represents a modest "offset" of the cost of prescription drugs. The CBO estimates that in 2008, average spending on prescription drugs among nonpoor (not low-income subsidy eligible) Medicare Part D participants

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was $1,800.[17] Note that given previous estimates of the effect of Part D on prescription drug use—for example Kaestner and Khan (2012) report an increase of 40 percent, or approximately $700—the offset related to inpatient care still yields a net increase in spending. Overall, the offset we estimate here represents a 3–5 percent of the total cost of prescription drugs among the elderly affected. If we assume that 11 million (28 percent) elderly gained prescription drug coverage through Medicare Part D, which is consistent with actual estimates, and that the offset of $47 (full sample) for inpatient Medicare payments applies, then the total "offset" is approximately $517 million, between $1 and $1.6 billion per year, or approximately 0.5 percent of the total annual cost of Medicare Part D.

In summary, our estimates suggest that gaining prescription drug insurance through Medicare Part D significantly reduces hospital admissions and hospital expenditures. The "offset" in downstream spending resulting from gaining prescription drug insurance is modest and represents approximately 0.5 percent of total Medicare Part D spending. The existence of such offsets lowers slightly the net cost of Medicare Part D—more so than estimated by the CBO. The existence of "offsets" also implies that stand-alone prescription drug plans such as those that characterize Medicare Part D cannot exploit these offsets through pricing because the "offsets" accrue to the medical insurer, which in this case is the federal government for fee-for-service Medicare enrollees. Medicare Advantage (HMO) plans can exploit this source of savings and this may partly explain why Medicare Advantage plans often included prescription drug benefits prior to Medicare Part D, as shown by Starc and Town (2015).

One limitation of our study is that our sample does not cover 100 percent of the Medicare population. For example, we exclude Medicare Advantage beneficiaries because there are no claims data for them in the MEDPAR. Our final sample covers approximately 65 percent of the Medicare population and includes fee-for-service beneficiaries who are not dual eligible, without ESRD or disability. Due to the sample limitation, it is unclear if our findings apply to, for example, Medicare Advantage beneficiaries.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of this article at the publisher's website.

Online Appendix Table 1. Holm–Bonferroni Adjusted P-Values for the Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on Hospital Admissions (Table 3)

Online Appendix Table 2. Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on Hospital Admissions (Unweighted)

Online Appendix Table 3. Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on Hospital Admissions Using Two-Period (Pre- and Post-Part D) Approach

Online Appendix Table 4. Placebo Tests: Reduced Form Estimates of the Effect of Prescription Drug Insurance on Hospital Admissions

Online Appendix Table 5. P-Values for Over-Identification Tests for Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on Hospital Admissions

Online Appendix Table 6. Reduced-Form (Event Study) Estimates of the Effect of Prescription Drug Insurance on Hospital Admissions

Online Appendix Table 7. Placebo Tests: Reduced Form Estimates of the Effect of Prescription Drug Insurance on Medicare Inpatient Payment

Online Appendix Table 8. P-Values of Over-Identification Tests for Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on Medicare Inpatient Payment

Online Appendix Table 9. Placebo Tests: Reduced Form Estimates of the Effect of Prescription Drug Insurance on Medicare Inpatient Charges

Online Appendix Table 10. P-Values of Over-Identification Tests for Instrumental Variables Estimates of the Effect of Prescription Drug Insurance on Medicare Inpatient Charges

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